Krishna Narayanan

CURRICULUM VITAE

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Research Interests

My current research focuses on questions arising in hardness of approximation, with broader interests in combinatorial optimization and computational complexity. My earlier work has included graph-theoretic problems and computational questions concerning polytopes and linear optimization.

Education

Ph.D., Mathematics and Computer Science Aug 2026–
University of Southern Denmark, Odense, Denmark
Advisor: Prof. Lars Rohwedder.

M.Sc., Mathematics Sep 2023–Dec 2025
Simon Fraser University, Burnaby, BC, Canada | CGPA: 3.85/4.33
Advisor: Prof. Tamon Stephen.
Thesis: *The Complexity of Monotone Shortest Paths on Simple Polytopes.*

Integrated M.Sc., Theoretical Computer Science Jun 2018–Jun 2023
PSG College of Technology, Coimbatore, India (via Anna University, Chennai, India)
Advisor: Prof. Lekshmi R S. | CGPA: 8.93/10; first class with distinction.

Publications and Manuscripts

Publications

[1] **Krishna Narayanan** and Tamon Stephen. *The Hardness of Monotone Eccentricity on Polytopes.* In: Misra, N., Pandey, A. (eds), *Algorithms and Discrete Applied Mathematics*, CALDAM 2026. Lecture Notes in Computer Science, vol. 16445. Springer, Cham. [DOI]

[2] Subhadeep R. Dev, Sanjana Dey, Florent Foucaud, **Krishna Narayanan**, and Lekshmi Ramasubramony Sulochana. *Monitoring edge-geodetic sets in graphs.* Discrete Applied Mathematics, vol. 377, 2025. [DOI]

[3] Florent Foucaud, **Krishna Narayanan**, and Lekshmi Ramasubramony Sulochana. *Monitoring edge-geodetic sets in graphs.* Algorithms and discrete applied mathematics, Lecture Notes in Comput. Sci., vol. 13947, Springer, Cham, 2023, pp. 245–256. MR 4573059. [DOI]

Preprints

Krishna Narayanan and Tamon Stephen. *A note on the exact partition polytope of Frieze and Teng.* [arXiv:2605.26505]

Work in Progress

Krishna Narayanan and Tamon Stephen. *The hardness of monotone eccentricity in simple polytopes and beyond.*

Awards and Fellowships

Graduate Fellowship, Simon Fraser University — Summer 2025

Graduate Travel and Research Award, Simon Fraser University — Spring 2025

MITACS Globalink Graduate Fellowship, MITACS, Canada — Jan 2024

Special Graduate Entrance Fellowship, Simon Fraser University, Canada — Sep 2023

Achievement Award, PSG Tech Alumni Association. Awarded for overall excellence over the years of study at PSG College of Technology, Coimbatore. — Mar 2023

Best Student Presentation, sponsored by Springer, for *Monitoring edge-geodetic sets in graphs*, 9th CALDAM, DA-IICT, Gandhinagar, India. — 2023

Joint Entrance Screening Test (JEST) — qualified; All India Rank 51. National Eligibility Test in Theoretical Computer Science. — 2023

Research Experience

Graduate Research and Teaching Assistant Sep 2023–Dec 2025
Simon Fraser University, BC, Canada
Research assistant under Prof. Tamon Stephen, focusing on questions in polytope theory. Teaching responsibilities are listed separately below.

Research Intern Jan 2023–Jun 2023
The Institute of Mathematical Sciences, Chennai, India
Advisor: Prof. Venkatesh Raman.
Studied algorithmic questions on *monitoring edge-geodetic sets in graphs*, with an emphasis on the complexity of the parameter in special graph classes; obtained partial results. Audited Parameterized Complexity and Approximation Algorithms.

MITACS Globalink Research Intern Jun 2022–Aug 2022
Memorial University of Newfoundland, St. John's, Canada
Advisors: Prof. Danny Dyer and Prof. Andrea Burgess.
Studied *the deduction number of graphs and subgraphs*, a cops-and-robber variant derived from the zero-visibility model. Investigated structural aspects and bounds, particularly on graph products.

Research Intern Apr 2021–Nov 2021
Université Clermont Auvergne, Clermont-Ferrand, France
Advisors: Prof. Florent Foucaud and Prof. Lekshmi R S.
Worked on *monitoring edge-geodetic sets in graphs*, a concept introduced for monitoring network vertices and/or edges to detect and prevent failures, inspired by edge-geodetic sets and distance-edge-monitoring sets.

Teaching Experience

Simon Fraser University — teaching assistantships 2023–2025
Calculus Workshop — office hours, invigilation and grading Fall 2025
MATH 232: Applied Linear Algebra — grading Summer 2025
MATH 308: Linear Optimization, Burnaby — tutorials and grading. Fall 2024 /
Introductory Math Workshop, Surrey — office hours and grading Spring 2025
Introductory Math Workshop, Surrey — office hours and grading Spring 2024
Algebra Workshop, Burnaby — office hours and grading Fall 2023

Selected Research Talks

The hardness of monotone eccentricity on polytopes Feb 2026
12th Annual International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Indian Institute of Technology Dharwad, Karnataka, India

Towards the computational complexity of monotone diameters of polytopes Oct 2025
West Coast Optimization Meeting, University of British Columbia, Okanagan Campus, Kelowna, BC, Canada

Monitoring edge-geodetic sets in graphs Oct 2024
Cascadia Combinatorial Feast, University of Victoria, Victoria, BC, Canada

Monitoring edge-geodetic sets in graphs Feb 2023
9th Annual International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), DA-IICT, Gandhinagar, India

Graph theoretical analysis of Buckminsterfullerene Feb 2020
Young Researchers' Forum, 6th Annual International Conference on Algorithms and Discrete Applied Mathematics (CALDAM), Indian Institute of Technology Hyderabad, India

Additional Experience

Curriculum Tech Intern, Mango Education Jun 2020–Sep 2020
Remote, Coimbatore, India. Manager: Mr. Arumugham Sankaran.
Curriculum development and analysis using graph-theoretic ideas.

Student Intern, C-DOT May 2019
Centre for Development of Telematics, Bangalore, India. Manager: Mr. Balamanikandan K.
Web application development for the Intellectual Property Rights and Knowledge Management Group.

Academic Service and Leadership

Councillor, Mathematics Graduate Caucus Jan 2025–Aug 2025
The Graduate Student Society, Simon Fraser University
Helped set the agenda, reviewed council packages, consulted caucus members to prepare questions and amendments, reported outcomes, and gathered members' input on upcoming business.

Secretary, Mathematics Graduate Caucus Sep 2023–Apr 2024
Simon Fraser University
Recorded and distributed general-meeting minutes in accordance with Graduate Students Society (GSS) standards and secured officer approval. Contributed to the Caucus's Cost of Living Survey addressing members' concerns about financial difficulties.

Executive Editor, The Bridge, PSG College of Technology's online student body magazine Jan 2020–Jun 2023

Joint Secretary, Astronomy Club, PSG College of Technology Oct 2020–Apr 2021

Member, Global Leaders Forum, PSG College of Technology. 2018–2023
Won multiple awards at school and collegiate Model United Nations (MUN) conferences. Authored an Economic Councils reference guide for MUNs, unveiled by Ambassador (Retd.) T.S. Tirumurthi, Indian Foreign Service, Govt. of India, at YLGC MUN 2024. Organizing team member, TEDxPSGTech 2022.

Co-ordinator, Editorial Team, COMAPP, a technical magazine of the Computational Sciences Association and Computer Applications Association, PSG College of Technology 2019/20; 2022/23

Selected Programs Attended

Certificate Program for Faculty in Computer Science Education — Advanced Graph Algorithms May 2023
Organized by the Indraprastha Institute of Information Technology, Delhi at SRM University, Chennai. Completed the module.

Indo-Dutch Pre-Conference School on Algorithms and Combinatorics, 9th CALDAM Feb 2023
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, India

ACM-India Winter School on Algorithms and Lower Bounds Jan 2022
Remote; organized by Chennai Mathematical Institute (CMI) and National Academy of Sciences (NASI)

Indo-French Pre-Conference School on Algorithms and Combinatorics, 6th CALDAM Feb 2020
Hosted and organized by Indian Institute of Technology Hyderabad, India

Technical Skills

Programming Languages — Past Familiarity: AMPL, Python, Java, Maple, SQL, Polymake.
Developer Tools: NetBeans, PyCharm.
Technologies/Frameworks: Linux, Windows, WordPress, Maple.

Interests

Self-declared "Av geek" (aviation geek). Formerly a competitive athlete and swimmer.